

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Electronics And Communication Engineering Department

Digital Electronics and Microprocessor & Microcontroller Laboratory Equipment

1. 8085 Microprocessor Trainer Kit

Overview

Overview:

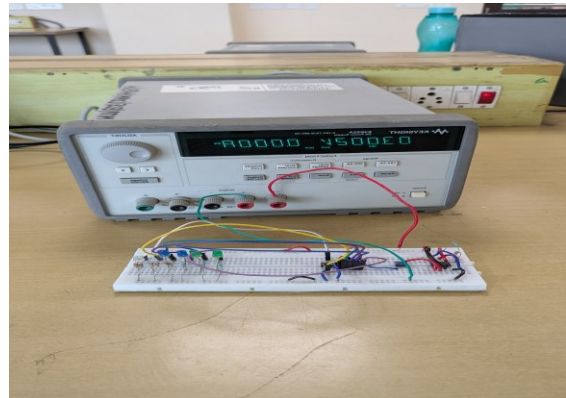
Provides hands-on training in 8085 microprocessor architecture, assembly language programming, memory organization, instruction execution, and peripheral interfacing.



2. Keysight DC Power Supply

Overview

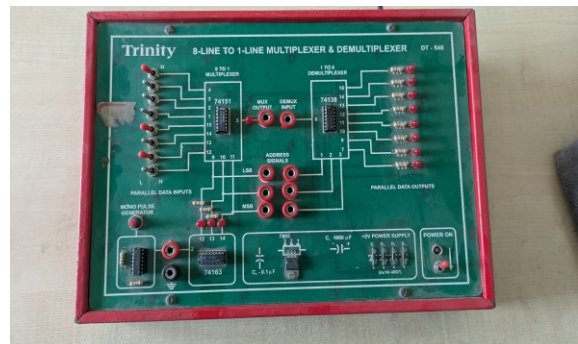
A precision regulated DC power supply that delivers stable voltage and current for powering electronic circuits and laboratory experiments.



3. 8:1 Multiplexer Trainer Kit

Overview

Designed to demonstrate the operation of an 8-to-1 multiplexer for data selection and digital signal routing.



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4. RS, D, T & JK Flip-Flop Trainer Kit

Overview

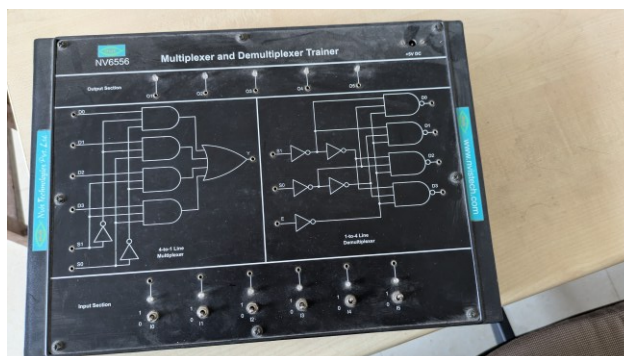
Helps students study various flip-flops, truth tables, timing diagrams, and their applications in sequential logic circuits.



5. MUX/DEMUX Trainer Kit

Overview

Used for understanding multiplexing and demultiplexing techniques employed in digital communication and switching systems.



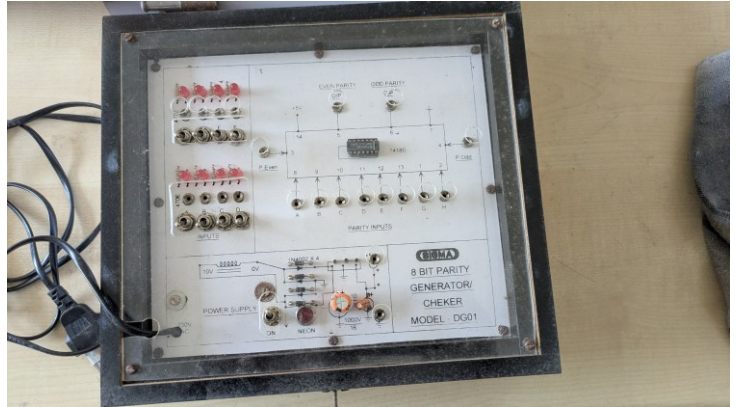
6. Parity Generator & Checker

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Overview

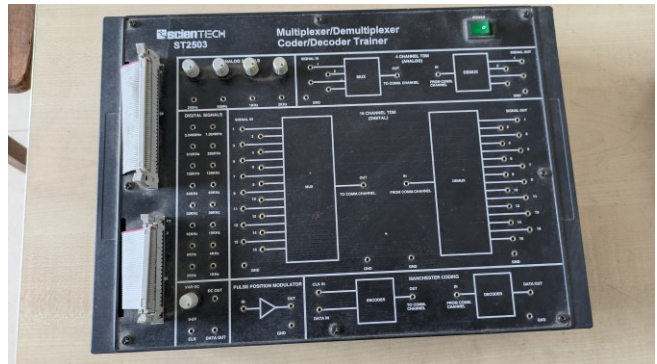
Demonstrates parity generation and parity checking techniques for error detection in digital data transmission.



7. 16:1 Multiplexer Trainer Kit

Overview

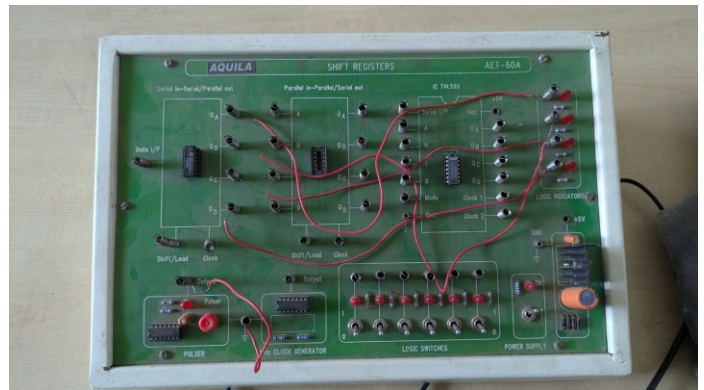
Provides practical knowledge of high-input multiplexers used in complex digital systems.



8. Shift Register Trainer Kit

Overview

Used to perform experiments on serial-to-serial, serial-to-parallel, and parallel data transfer operations.



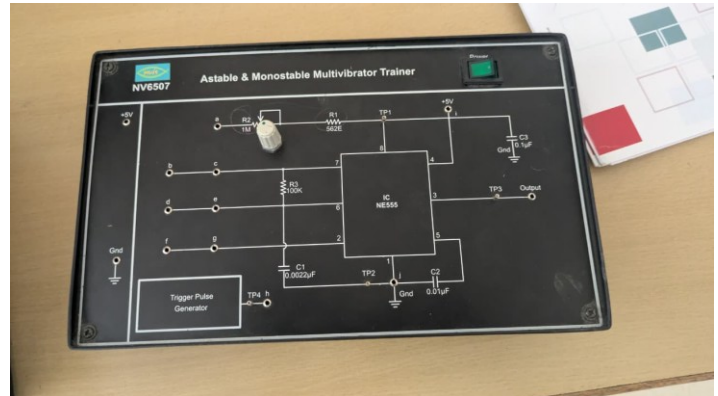
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9. Astable Multivibrator Trainer

Overview

Generates continuous square-wave pulses used as timing and clock signals in digital circuits.



10. Desktop Computer

Overview

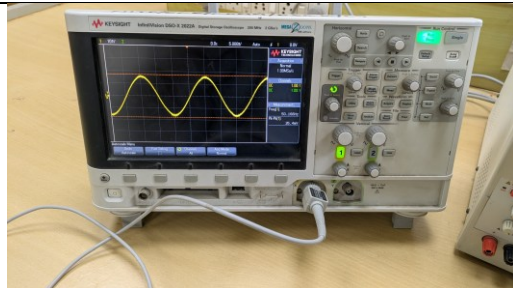
Used for programming, simulation, embedded software development, and interfacing with laboratory hardware.



11. Digital Storage Oscilloscope (DSO)

Overview

Displays and analyzes electrical waveforms for testing, measurement, debugging, and troubleshooting electronic circuits.



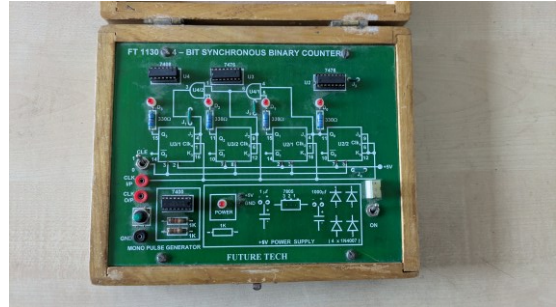
12. Binary Counter Trainer

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Overview

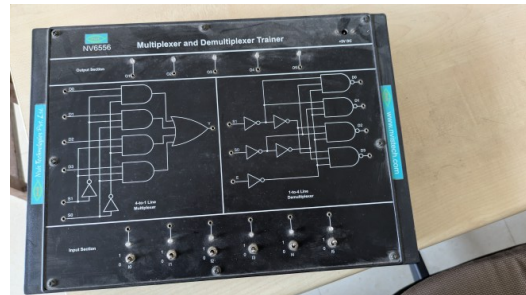
Demonstrates the operation of synchronous and asynchronous binary counters used in digital systems.



13. 4:1 Multiplexer Trainer

Overview

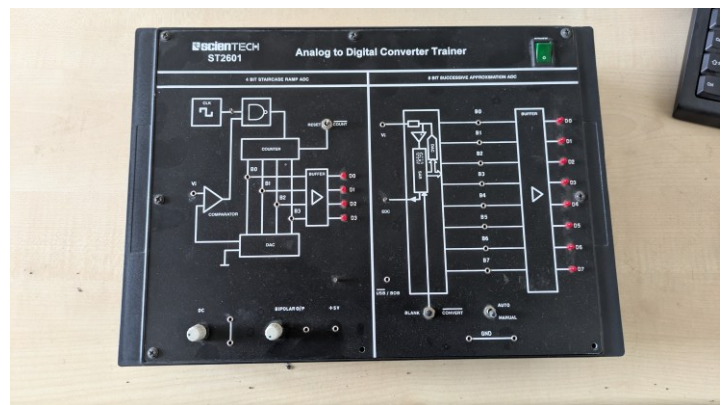
Introduces the working principles and practical implementation of a 4-to-1 multiplexer.



14. Analog-to-Digital Converter (ADC) Trainer

Overview

Demonstrates the conversion of analog input signals into digital data for embedded and data acquisition systems.



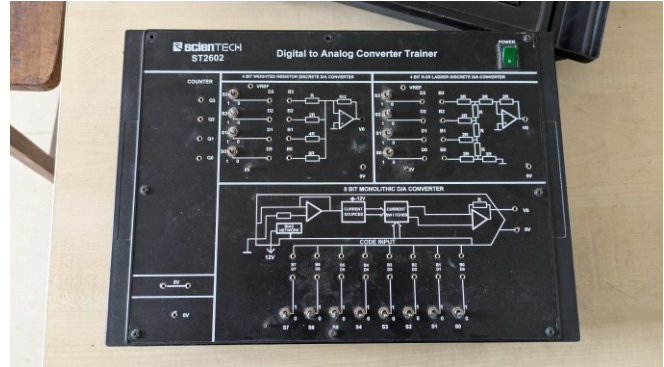
15. Digital-to-Analog Converter (DAC) Trainer

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Overview

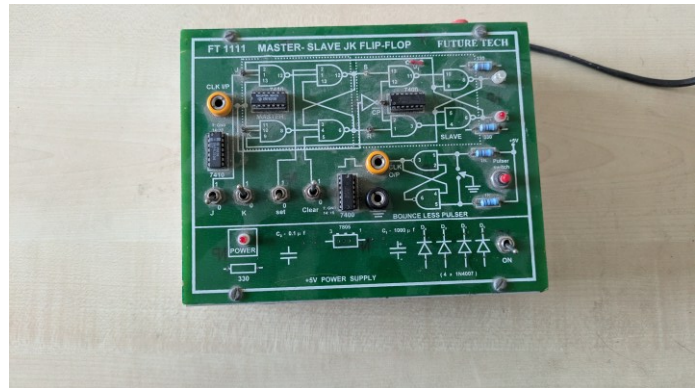
Converts digital data into analog output signals used in control and signal processing applications.



16. Master-Slave JK Flip-Flop Trainer Kit

Overview

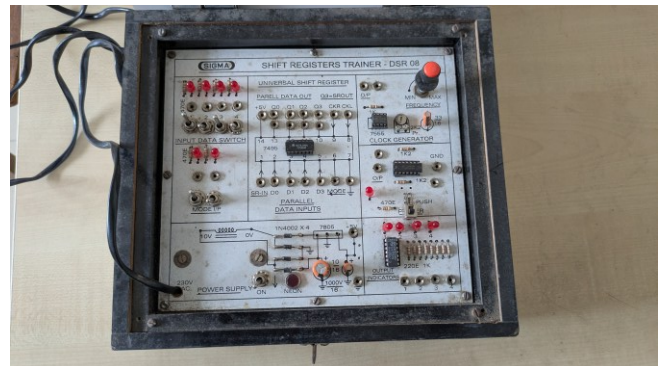
Provides practical understanding of master-slave JK flip-flop operation and edge-triggered sequential circuits.



17. Shift Register Trainer (DSR-08)

Overview

Supports advanced experiments on serial and parallel data communication using shift registers.



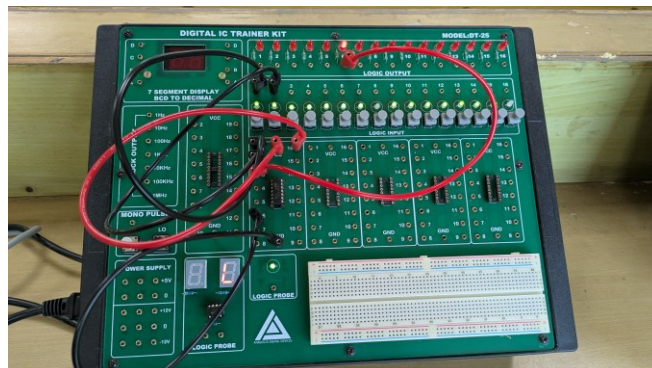
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18. Digital IC Trainer Kit

Overview

A versatile platform for testing and experimenting with standard digital integrated circuits.



19. 8086 Microprocessor Trainer Kit

Overview

Provides practical experience with 16-bit microprocessor architecture, assembly programming, memory management, and interfacing.



20. 8051 Microcontroller Trainer Kit

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Overview

Enables embedded system development through programming, interfacing, and hardware control using the 8051 microcontroller.



21. Universal Flash Programmer (LT-1270)

Overview

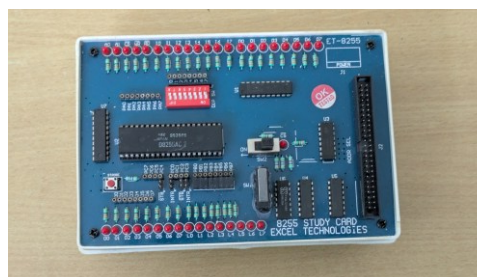
Used to program and verify microcontrollers, Flash memories, EEPROMs, and programmable integrated circuits.



22. 8255 Interface Card

Overview

Provides programmable parallel input/output interfacing between microprocessors and external devices.



23. 8279 Interface Card

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Overview

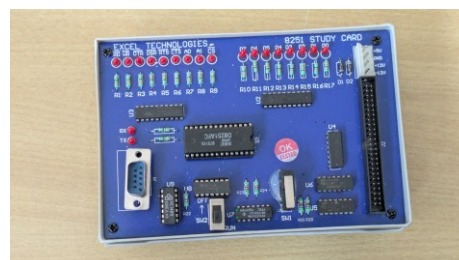
Designed for keyboard scanning and display interfacing applications in microprocessor systems.



24. 8251 Interface Card

Overview

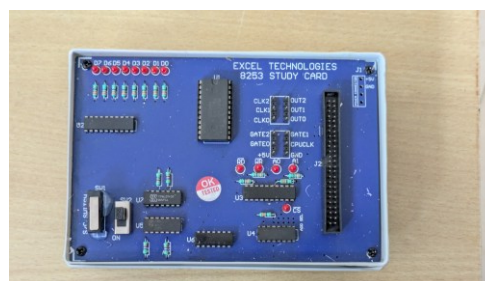
Demonstrates serial communication using the Universal Synchronous/Asynchronous Receiver/Transmitter (USART).



25. 8253 Interface Card

Overview

Provides programmable timer and counter functions for timing and frequency measurement applications.



26. DMA-8257 Interface Card

Overview

Demonstrates Direct Memory Access (DMA) for efficient high-speed data transfer without CPU intervention.



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27. Stepper Motor Interfacing Card	
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Overview	
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Used to control and interface stepper motors for embedded control and automation experiments.

28. LED Display Matrix Card	
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Overview	
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Demonstrates programming and interfacing of LED matrix displays for text and graphical output.

29. ARM Processor Development Kit	
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Overview	
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Supports advanced embedded system programming and real-time application development using ARM processors.

30. PID Controller (ST2451)	

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Overview

Demonstrates closed-loop process control using Proportional, Integral, and Derivative (PID) control techniques.



31. PLC Controller (ST2403A)

Overview

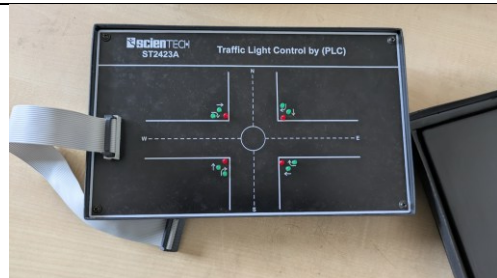
Provides practical training in Programmable Logic Controller (PLC) programming and industrial automation.



32. Traffic Light Controller by PLC (ST2423A)

Overview

Simulates traffic signal control using PLC programming for automation and sequence control.



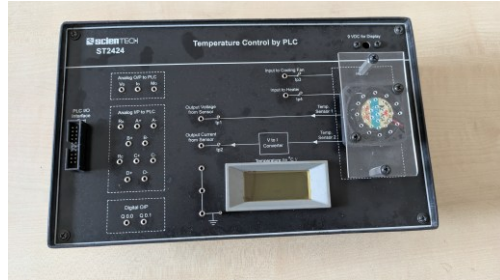
33. Temperature Controller by PLC (ST2424)

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Overview

Demonstrates automatic temperature monitoring and process control using PLC-based systems.



34. Elevator Controller by PLC (ST2422)

Overview

Overview:
Illustrates elevator sequencing, safety logic, and control using PLC programming for industrial automation.

